

Paper 3, Theorem 2.2: Statement Amendment

Internal research note for the **ShenWork** formalization

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Abstract

The nonlinear stability clause printed in Theorem 2.2 of Chen–Ruau–Shen, Part II, asserts a C^1 exponential estimate valid for *all* $t \geq 0$ from initial data small only in L^∞ . Under the theorem’s own initial-data hypothesis this clause is not well posed at $t = 0$: the C^1 norm need not be finite, and an L^∞ ball contains functions of arbitrarily large C^1 norm. The argument in fact proves two correct statements—an all-time estimate in the fractional-power (strong) norm and an *eventual* C^1 estimate for L^∞ -small data. This note records the explicit counterexample to the printed clause, locates the gap, gives the corrected two-stage formulation, and states the corresponding Lean obstructions. This is the second statement-level issue found while formalizing the series; the first is the parameter error in Theorem 1.2 of Part I (companion note).

1 Source checked

The source examined is the original submission, not a repository paraphrase:

Le Chen, Ian Ruau, and Wenxian Shen, *Chemotaxis models with signal-dependent sensitivity and a logistic-type source, II: Persistence and stabilization*, arXiv:2604.02599v1, 3 April 2026.

Theorem 2.2 there is titled “Linear stability and instability.” For the positive constant equilibrium (u_*, v_*) with $u_* = (a/b)^{1/\alpha}$ and $a, b > 0$, part (1) states:

- (i) if $\chi_0 < \chi_{a,b,\beta}^*(u_*)$ the equilibrium is linearly stable;
- (ii) if $\chi_0 > \chi^*$ it is linearly unstable;
- (iii) moreover, in the stable regime there exist $C, \delta, \lambda > 0$ such that every u_0 satisfying the initial-data condition (1.8) *and* a smallness condition obeys the exponential estimate (2.12) *for all* $t \geq 0$ in the C^1 norm.

The initial-data hypothesis (1.8) requires only $u_0 \in C(\overline{\Omega})$, positive. Proposition 1.1 provides convergence to u_0 in L^∞ as $t \rightarrow 0^+$ and classical spatial regularity only for *positive* time. There is no C^1 , Sobolev, or fractional-power smallness hypothesis on u_0 in the statement of Theorem 2.2.

2 The over-statement in the printed (2.12)

The printed nonlinear clause asserts, for data small only in L^∞ , a C^1 exponential estimate holding at *every* $t \geq 0$, including $t = 0$:

$$\|u(t) - u_*\|_{C^1(\overline{\Omega})} \leq C e^{-\delta t} \|u_0 - u_*\|_*, \quad t \geq 0.$$

This is not well posed under the theorem’s own hypothesis:

- u_0 need only lie in $C(\overline{\Omega})$, so $\|u_0 - u_*\|_{C^1}$ may be infinite and the $t = 0$ instance meaningless;
- even for C^1 data, an L^∞ ball contains functions of arbitrarily large C^1 norm, so no single C controls the $t = 0$ (or $t \rightarrow 0^+$) C^1 size uniformly over the admitted class.

Explicit counterexample. On the unit interval $\Omega = (0, 1)$ with Neumann conditions, take

$$u_{0,N}(x) = u_* + N^{-1/2} \cos(N\pi x), \quad N \in \mathbb{N}.$$

For large N this is positive and satisfies (1.8), with

$$\|u_{0,N} - u_*\|_{L^\infty} = N^{-1/2} \rightarrow 0, \quad \|u_{0,N} - u_*\|_{C^1} = N^{1/2}\pi \rightarrow \infty$$

(and, in the minimal model, exact mass u_*). Hence no uniform-in-data $t = 0$ estimate of the form (2.12) can hold: the smallness is measured in L^∞ , the conclusion in C^1 , and the two are unrelated at $t = 0$.

3 What the published proof actually establishes

Section 5.1 proves the result in two distinct stages, both correct.

Stage A (strong norm, all time). For data small in the fractional-power space $X^\alpha = D((I - L)^\alpha)$, which in one dimension embeds into C^1 for $\alpha > 3/4$, the standard sectorial nonlinear-stability theorem (Henry, *Geometric Theory of Semilinear Parabolic Equations*, Ch. 5–6) gives exponential decay in X^α for all $t \geq 0$. This is instantaneous and correct.

Stage B (L^∞ -small, eventual). For data small only in L^∞ , the proof first evolves to a *positive* time t_0 , at which analytic-semigroup smoothing places the solution in a small X^α neighborhood, and then applies Stage A. This yields exponential decay for $t \geq t_0$ —an *eventual* estimate, not a $t = 0$ estimate.

The mechanism proves an all-time estimate in the strong norm and an eventual estimate in C^1 for L^∞ -small data. It does not prove the printed (2.12) at $t = 0$ from L^∞ data; that conjunction is the over-statement.

4 Corrected statement

Split the printed clause into the two statements the proof supports.

- (1) **Strong norm, all time.** There exist $\alpha \in (3/4, 1)$ and $\rho, C, \delta > 0$ such that

$$\|u_0 - u_*\|_{X^\alpha} \leq \rho \implies \|u(t) - u_*\|_{X^\alpha} \leq C e^{-\delta t} \|u_0 - u_*\|_{X^\alpha} \quad \text{for all } t \geq 0.$$

- (2) **L^∞ -small, eventual.** There exist $\varepsilon, C', \delta' > 0$ and, for each admissible datum, a time $t_0 = t_0(u_0) > 0$ such that

$$\|u_0 - u_*\|_{L^\infty} \leq \varepsilon \implies \|u(t) - u_*\|_{C^1} \leq C' e^{-\delta'(t-t_0)} \quad \text{for all } t \geq t_0.$$

The linear stability/instability dichotomy of Theorem 2.2 is unaffected and correct; only the nonlinear clause (2.12) needs the $t = 0$ -versus-eventual and L^∞ -versus- X^α distinctions made explicit.

5 A companion parameter-fidelity point

The sufficient stability bound $\kappa < (\sqrt{\mu} + \sqrt{a\alpha})^2$, with $\kappa = \chi_0 \gamma \nu u_*^\gamma (1 + v_*)^{-\beta}$, is the *continuous* minimum of $(\lambda + \mu)(\lambda + a\alpha)/\lambda$ over $\lambda > 0$. On a bounded domain the Neumann spectrum is discrete ($\lambda_n = n^2 \pi^2$ on $(0, 1)$), so the exact stable half of Theorem 2.2 is governed by the discrete infimum $\min_{n \geq 1} \sigma_n < 0$, which the continuous bound only implies and can be strictly stronger than needed. A faithful formalization should clear the discrete mode condition, not stop at the continuous sufficient bound.

6 Lean amendment

The repository records two formal obstructions to the literal clause, both **sorry-free**.

- `Paper3.not_SectorialLocalExponentialRaw_constant_c1Distance` (`ShenWork/Paper3/Statements.lean`) proves the raw sectorial local-exponential branch false when the C^1 distance is unrelated to the dynamics: a constantly-1 C^1 distance would force $2 \leq Ce^{-\text{rate}t}$ for all t , impossible as the right side tends to 0.
- `ShenWork/Paper3/IntervalDomainSectorialCorrectedObstruction.lean` records a *zero-time* obstruction: `InitialTrace` controls only the $t \rightarrow 0^+$ deleted limit, not the stored slice $u(0)$, so a global solution may be re-anchored to an arbitrarily large constant slice at $t = 0$ without changing global classical solvability or the initial trace—hence no all-time $t = 0$ C^1 bound can hold.

The corrected target used in the repository is the eventual, equilibrium-specific, full-mode orbit bound `IntervalDomainSpectralSemigroupOrbitBoundEventualEquilibriumWithoutMass` (linearized multiplier $e^{-d_k t}$ with $d_k = a\alpha + \lambda_k - \kappa \lambda_k / (\lambda_k + \mu)$ for every mode k including $k = 0$, and a uniform positive delay t_0), together with `LocallyExponentiallyStableFromSup` for the positive equilibrium (the mass constraint is needed only in the minimal $a = b = 0$ model). The over-stated `...Raw` frontier—the pure Neumann heat multiplier quantified over all $t \geq 0$ —is not the correct analytic object and is refuted for a genuine C^1 distance from L^∞ data.

7 Relation to Part I (arXiv:2512.14858) Theorem 1.2

This is the second statement-level issue found while formalizing the Chen–Ruau–Shen series. The first (companion note) is a genuine *parameter* error: Theorem 1.2 of Part I omits the guard excluding $a > 0$, $b = 0$, a branch in which the solution is provably unbounded ($M'(t) = aM(t)$). The present issue is a *statement over-statement*: the intended stability result of Part II holds in the corrected two-stage form above, but the printed (2.12)—a $t = 0$ C^1 estimate from L^∞ -small data—is not well posed as written.